

- 1 (a) A body has an initial velocity u and an acceleration a . After a time t , the body has moved a displacement s and has a final velocity v . One of the equations of motion of this body is

$$s = ut + \frac{1}{2}at^2$$

State the conditions that must be satisfied for the above equation to be valid.

Conditions for equations to be valid:

- Constant acceleration [B1]
- Motion in a straight line [B1]

[2]

- (b) A hot air balloon was moving at a velocity of 11.7 m s^{-1} , at an angle of 59.0° from the horizontal, as shown in Fig. 1.1 below.

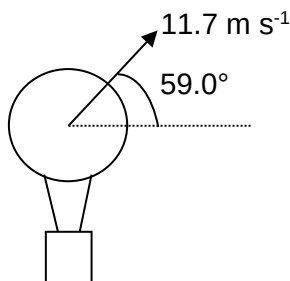


Fig. 1.1

- (i) Determine the vertical component of the velocity of the balloon.

$$\sin \theta = \frac{v_y}{v}$$

$$v_y = v \sin \theta = 11.7 \sin(59.0) = 10.0 \text{ m s}^{-1}$$

vertical component of the velocity = m s^{-1} [1]

- (ii) A slotted mass is released from the balloon. Fig. 1.2 shows the subsequent path of the slotted mass. The dotted figure shows the position of the hot air balloon at the instant when the slotted mass is released.

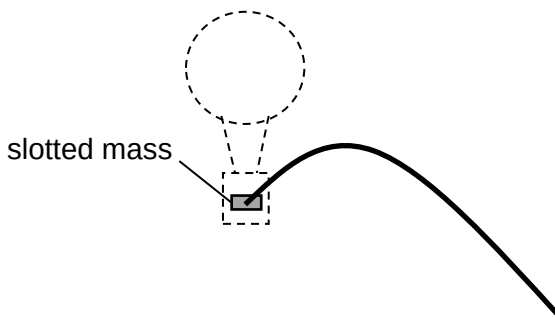


Fig. 1.2

1. Throughout the motion, the slotted mass is observed to be directly below the hot air balloon. Explain why this is so.

Both the mass and balloon have the same horizontal velocity [B1], hence their horizontal displacements are the same.

[1]

2. Determine how far below the balloon would the slotted mass be after 3.0 s. You may assume that the slotted mass has not yet landed on the ground and that air resistance on the slotted mass is negligible.

$$\begin{aligned}
 s_{\text{mass}} &= 10.0 (3.0) + \frac{1}{2}(-9.81)(3.0)^2 \\
 &= -14.1 \text{ m} & [\text{B1}] \\
 s_{\text{balloon}} &= 10.0 (3.0) = 30.0 \text{ m} & [\text{M1}] \\
 \text{Distance} &= 30.0 - (-14.1) = 44.1 \approx 44 \text{ m} & [\text{A1}]
 \end{aligned}$$

distance = m [3]

3. Describe qualitatively the changes, if any, to the answer in **(b)(ii)2** if a 100 kg cargo was dropped from the balloon instead of the slotted mass. Assume air resistance on the cargo is negligible too.

The distance between cargo and balloon will be greater.

.....[1]

[Total: 8]